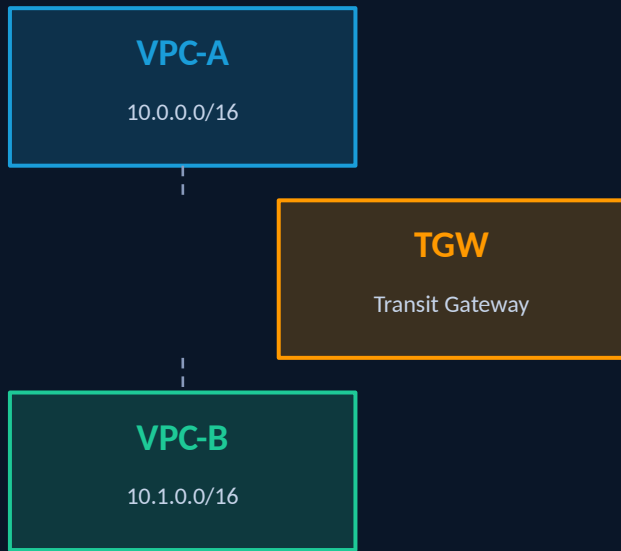


AWS

ADVANCED NETWORKING

Enabled Multi-VPC Communication using AWS Transit Gateway

Hub-and-Spoke Architecture • Cross-VPC Private Routing • Enterprise Networking



Project Overview

Problem Statement

- VPCs are isolated by default — no native cross-VPC communication
- VPC Peering scales poorly: $N*(N-1)/2$ connections for N VPCs
- Complex route table management across peered VPCs
- No centralized control for inter-VPC traffic

Objectives

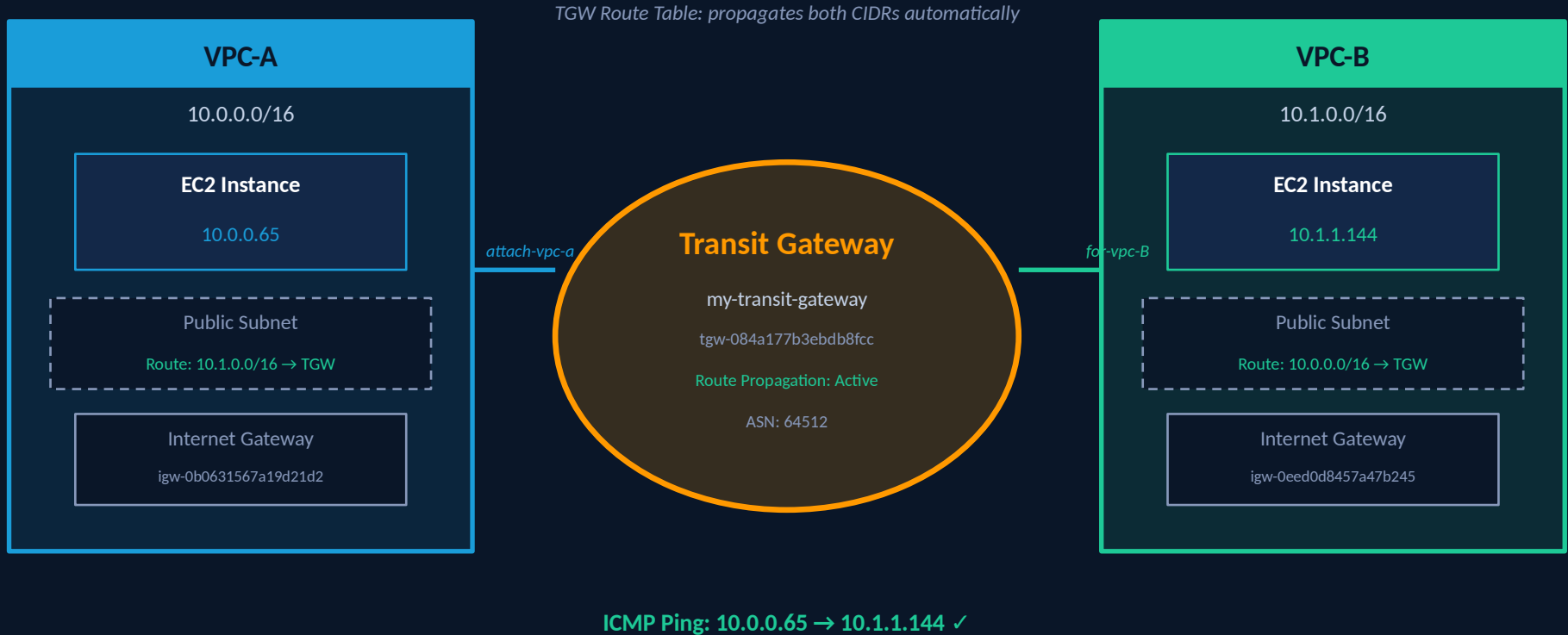
- Deploy a scalable Transit Gateway as central routing hub
- Attach VPC-A (10.0.0.0/16) and VPC-B (10.1.0.0/16)
- Configure propagated routes for automatic learning
- Validate private IP connectivity via ICMP ping

Expected Outcome

- Private communication between isolated VPCs
- Hub-and-spoke model ready for enterprise scale
- Simplified routing — single TGW manages all paths
- Zero internet exposure — fully private traffic

Architecture Overview

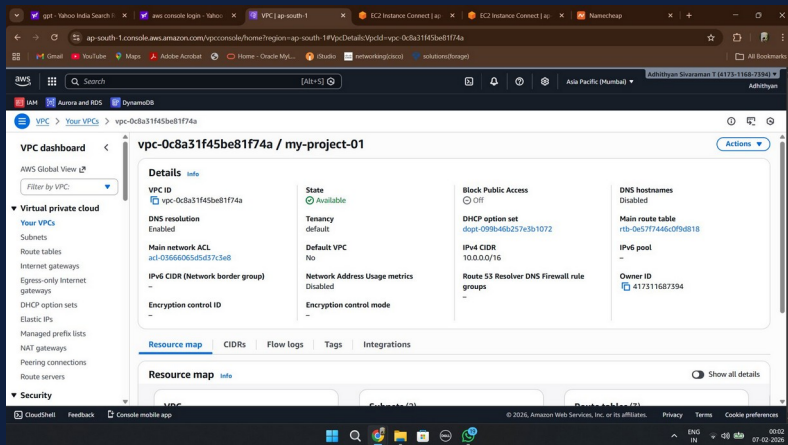
Hub-and-Spoke • AWS ap-south-1



Environment Setup

Two isolated VPCs — non-overlapping CIDR blocks

VPC-A / my-project-01

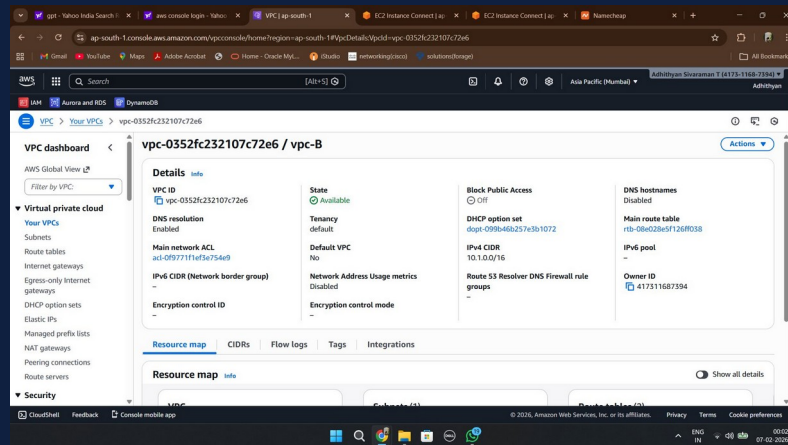


CIDR: 10.0.0.0/16

State: Available ✓

Subnets: 2 public subnets

VPC-B



CIDR: 10.1.0.0/16

State: Available ✓

Subnets: 1 public subnet

Both VPCs are isolated — no direct routing exists without Transit Gateway

Transit Gateway Deployment

Centralized routing hub — one gateway to connect all VPCs

The screenshot shows the AWS Management Console interface for a Transit Gateway. The breadcrumb navigation indicates the path: VPC > Transit gateways > tgw-084a177b3ebdb8fcc. The main content area displays the details for the gateway 'my-transit-gateway' (ID: tgw-084a177b3ebdb8fcc). The details are organized into a grid with the following information:

Details			
Transit gateway ID tgw-084a177b3ebdb8fcc	State Available	Amazon ASN 64512	DNS support Enable
Transit gateway ARN arn:aws:ec2:ap-south-1:417311687394:transit-gateway/tgw-084a177b3ebdb8fcc	Default association route table Enable	Association route table ID tgw-rtb-0b48a7ea8a66e841f	Auto accept shared attachments Disable
Owner ID 417311687394	Default propagation route table Enable	Propagation route table ID tgw-rtb-0b48a7ea8a66e841f	VPN ECMP support Enable
Description -	Transit gateway CIDR blocks -	Security Group Referencing support Disable	Metering policy ID -
Multicast support Disable	Encryption support Disabled	Encryption state message -	

Below the details grid, there are tabs for 'Flow logs', 'Sharing', and 'Tags'. The 'Flow logs' tab is currently selected.

TGW Details

Name	my-transit-gateway
State	Available ✓
ASN	64512
DNS Support	Enabled
Route Assoc.	Enabled
Route Prop.	Enabled
Encryption	Disabled
Multicast	Disabled

💡 Default propagation enables automatic route learning — no manual static routes needed

Hub-and-Spoke model • Scales to 5,000 VPC attachments • Cross-account support

Transit Gateway Attachments

Both VPCs attached and associated to TGW route table

Transit gateway attachments (1/2)

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID
☒ attach-vpc-a	tgw-attach-02c47b1f7708b7fd5	tgw-084a177b3ebdb8fcc	Available	VPC	vpc-0c8a31f45be81f74a
☐ for-vpc-B	tgw-attach-0a36211c7feda93f6	tgw-084a177b3ebdb8fcc	Available	VPC	vpc-0352fc232107c72e6

Transit gateway attachment: tgw-attach-02c47b1f7708b7fd5 / attach-vpc-a

Details

Transit gateway attachment ID tgw-attach-02c47b1f7708b7fd5	State Available	Resource type VPC	Association state Associated
Transit gateway ID tgw-084a177b3ebdb8fcc	Resource owner ID 417311687394	Resource ID vpc-0c8a31f45be81f74a	Association route table ID tgw-rtb-0b48a7ea8a66e841f

attach-vpc-a

ID: tgw-attach-02c47b1f7708b7fd5

VPC: vpc-0c8a31f45be81f74a

State: Available • Associated

for-vpc-B

ID: tgw-attach-0a36211c7feda93f6

VPC: vpc-0352fc232107c72e6

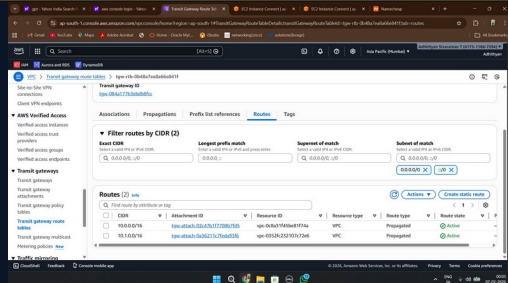
State: Available • Associated

Resource type: VPC • Association state: Associated • TGW: tgw-084a177b3ebdb8fcc

Route Configuration

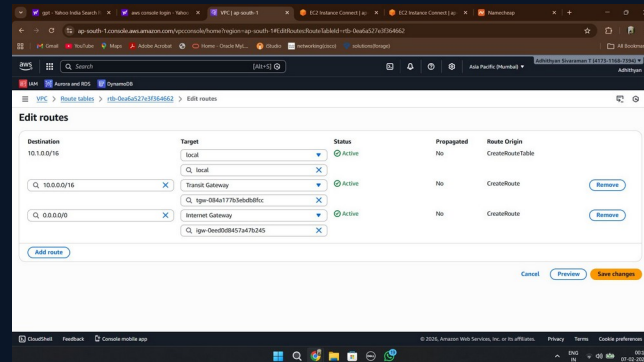
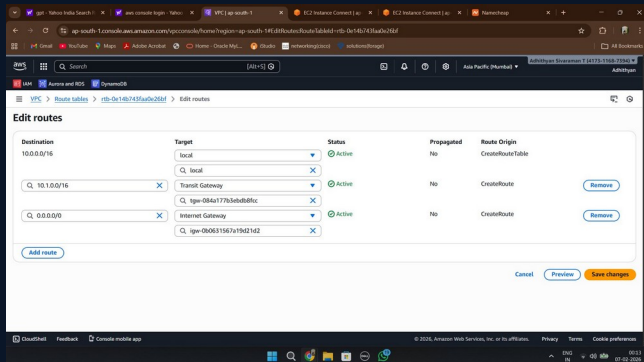
Three-tier routing: TGW Route Table + VPC-A Routes + VPC-B Routes

Transit Gateway Route Table (tgw-rtb-0b48a7ea8a66e841f) — 2 Propagated Routes



VPC-A Route Table (rtb-0e14b743faa0e26bf)

VPC-B Route Table (rtb-0ea6a527e3f364662)



All routes propagated — no static entries required • Dynamic route learning via TGW

Compute Layer

EC2 instances deployed in each VPC as traffic endpoints

The screenshot shows the AWS Management Console for the 'Instances' page. The table lists the following instances:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
proj-3-instance	i-0b51ed05a19290404	Stopped	t2.micro	-	view alarms +	ap-south-1a	-
vpc-a-instance	i-042855a2f4ec88a1b	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a	-
instance-for-B	i-02e68aa20881acdc9	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a	-
project5-webs...	i-08efeb8bcd62c7e7	Stopped	t2.micro	-	View alarms +	ap-south-1b	-

Below the table, the '2 instances selected' section shows performance metrics for the selected instances:

- CPU utilization (%): 0.845% (vpc-a-instance), 0.422% (instance-for-B)
- Network in (bytes): 15.55K (vpc-a-instance), 7.775K (instance-for-B)
- Network out (bytes): 14.25K (vpc-a-instance), 7.128K (instance-for-B)
- Network packets in (count): 154.66 (vpc-a-instance), 77.33 (instance-for-B)

vpc-a-instance

ID: i-042855a2f4ec88a1b

Type: t2.micro

Private IP: 10.0.0.65

State: Running ✓

instance-for-B

ID: i-02e68aa20881acdc9

Type: t2.micro

Private IP: 10.1.1.144

State: Running ✓

Amazon Linux 2023

t2.micro

2/2 Checks Passed

ap-south-1a

Private Routing

Instances communicate exclusively via private IPs — no public IP routing involved

Security Controls

Layered security: Security Groups • NACLs • Private routing

The screenshot displays the AWS Management Console interface for a Security Group. The left-hand navigation pane includes sections for Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The 'Network & Security' section is expanded, showing 'Security Groups'. The main content area displays the details for the Security Group 'sg-03e228d77453c1d90'. The 'Inbound rules' tab is selected, showing a table of three rules: HTTP (port 80), All ICMP (all ports), and SSH (port 22). The 'Outbound rules' tab shows one rule for SSH (port 22). The 'VPC associations' tab shows the VPC ID 'vpc-0c8a31f45be81f74a'. The 'Related resources' tab shows the launch wizard-2 created on 2026-02-06T17:11:30.847Z.

Type	Protocol	Port	Source
HTTP	TCP	80	0.0.0.0/0
All ICMP	ICMP	All	0.0.0.0/0
SSH	TCP	22	0.0.0.0/0

security-group-for-A

Type	Protocol	Port	Source
HTTP	TCP	80	0.0.0.0/0
All ICMP	ICMP	All	0.0.0.0/0
SSH	TCP	22	0.0.0.0/0

ICMP Enabled:

Allows cross-VPC ping for connectivity testing

SSH Access:

Port 22 open for EC2 Instance Connect

HTTP Ready:

Port 80 for application traffic (future use)

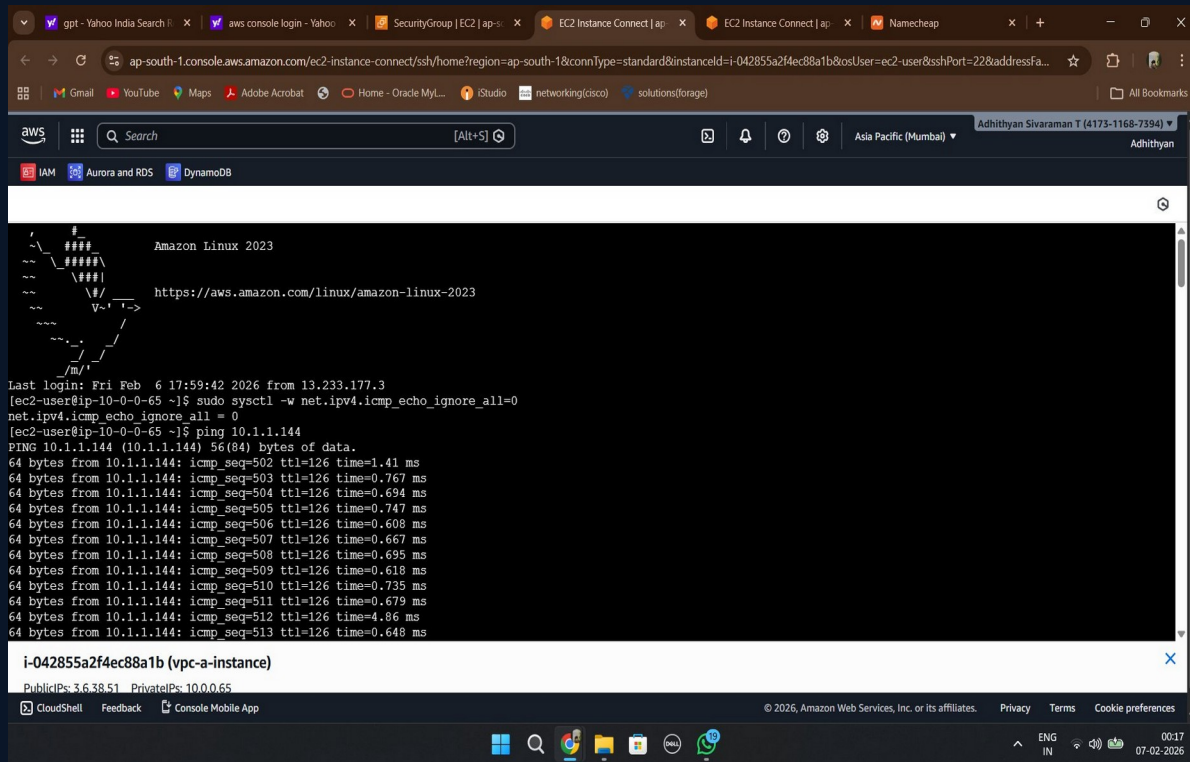
Network Isolation:

VPCs communicate only via TGW — no direct peering

Connectivity Validation

✓ VALIDATED

Cross-VPC ICMP ping — 10.0.0.65 → 10.1.1.144



```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Fri Feb  6 17:59:42 2026 from 13.233.177.3
[ec2-user@ip-10-0-0-65 ~]$ sudo sysctl -w net.ipv4.icmp_echo_ignore_all=0
net.ipv4.icmp_echo_ignore_all = 0
[ec2-user@ip-10-0-0-65 ~]$ ping 10.1.1.144
PING 10.1.1.144 (10.1.1.144) 56(84) bytes of data:
64 bytes from 10.1.1.144: icmp_seq=502 ttl=126 time=1.41 ms
64 bytes from 10.1.1.144: icmp_seq=503 ttl=126 time=0.767 ms
64 bytes from 10.1.1.144: icmp_seq=504 ttl=126 time=0.694 ms
64 bytes from 10.1.1.144: icmp_seq=505 ttl=126 time=0.747 ms
64 bytes from 10.1.1.144: icmp_seq=506 ttl=126 time=0.608 ms
64 bytes from 10.1.1.144: icmp_seq=507 ttl=126 time=0.667 ms
64 bytes from 10.1.1.144: icmp_seq=508 ttl=126 time=0.695 ms
64 bytes from 10.1.1.144: icmp_seq=509 ttl=126 time=0.618 ms
64 bytes from 10.1.1.144: icmp_seq=510 ttl=126 time=0.735 ms
64 bytes from 10.1.1.144: icmp_seq=511 ttl=126 time=0.679 ms
64 bytes from 10.1.1.144: icmp_seq=512 ttl=126 time=4.86 ms
64 bytes from 10.1.1.144: icmp_seq=513 ttl=126 time=0.648 ms

i-042855a2f4ec88a1b (vpc-a-instance)
PublicIPs: 3.6.38.51 PrivateIPs: 10.0.0.65
```

Source

10.0.0.65

VPC-A EC2

Destination

10.1.1.144

VPC-B EC2

Packet Size

64 bytes

56(84) bytes

TTL

126

Expected cross-VPC

Latency

< 1ms

Same-region routing

Packet Loss

0%

Stable connection

✓ Private IP communication confirmed • Zero packet loss • Sub-millisecond latency

Enterprise Benefits

Why Transit Gateway over VPC Peering for enterprise environments



Hub-and-Spoke Architecture

Single TGW as central hub — connect dozens of VPCs without mesh peering complexity



Centralized Routing

One route table to manage all inter-VPC paths — consistent policy enforcement



Reduced Peering Complexity

N VPCs = N attachments (not $N * (N-1) / 2$ peering connections)



Dynamic Route Learning

Route propagation auto-syncs routing tables — zero manual static route management



Scalable Connectivity

Supports up to 5,000 VPC attachments — ready for enterprise multi-account setups



Production-Ready Design

Foundation for Direct Connect, Site-to-Site VPN, and shared services integration

Project Outcomes & Future Enhancements

✓ Project Outcomes

- Created Two Isolated VPCs with non-overlapping CIDRs
- Deployed Transit Gateway as centralized routing hub
- Configured TGW Attachments for both VPCs
- Enabled dynamic route propagation across attachments
- Configured VPC route tables pointing to TGW
- Validated private cross-VPC communication via ICMP
- Demonstrated Hub-and-Spoke architecture pattern



Future Enhancements

Multi-Region TGW Peering

Connect VPCs across AWS regions

Site-to-Site VPN

Extend to on-premises data centers

AWS Direct Connect

Dedicated private connectivity

Shared Services VPC

Centralize DNS, NTP, monitoring

AWS Network Firewall

Centralized traffic inspection

Multi-Account Setup

Resource Access Manager integration

TGW Flow Logs

Advanced traffic monitoring & audit



Production-ready design • [GitHub Portfolio](#) • [LinkedIn Showcase](#) • Interview Ready